

EUN-SEO HAN

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EDUCATION

Myongji University

Seoul, Korea

M.Eng., Transportation Engineering

Sept. 2023 – Feb. 2025

- GPA: 4.25/4.5, Advisor: Prof. Ho-Chul Park
- Thesis: "Efficiency Evaluation of Metropolitan Subway Transfer Stations Focusing on Integrated Transfers between Transportation Modes"
- Teaching Assistant, "Traffic Capacity" (Fall 2023, Fall 2024); "Urban Traffic Operations" (Spring 2024)

B.Eng., Transportation Engineering

Mar. 2020 – Aug. 2023

- CGPA: 4.16/4.5, Major GPA: 4.23/4.5

RESEARCH INTERESTS

Transportation System & Operations: Travel behavior, Service performance evaluation, Transfer & connectivity analysis

Public Transit: Intermodal connectivity, Shared & micromobility, Mobility-as-a-Service (Maas), Smart mobility

Data Analytics: Smartcard, Vehicle trajectory data analysis

RESEARCH EXPERIENCE

The Seoul Institute

Seoul, Korea

Researcher, Smart Transportation Research Division

Mar. 2025 – Present

Research Assistant, Smart Transportation Research Division

Jan. 2025 – Feb. 2025

"Feasibility Reassessment and Development Strategy for Seoul Urban Railway Network Plan"

Seoul Metropolitan Government | Jul. 2025 – Present

- Analyze travel demand patterns, network connectivity, and accessibility impacts to assess feasibility of proposed and petitioned rail lines.
- Identify priority corridors by incorporating inputs from district governments and public stakeholders.
- Design a Seoul Urban Railway Network Plan aligned with travel behavior and policy goals, reducing underserved rail service areas from 37% to 27%.

Sustainable Urban Transport System for Carbon Neutrality in Bishkek, Kyrgyzstan

Korea International Cooperation Agency (KOICA) | Jan. 2025 – Jun. 2025

- Evaluated urban transport conditions using multi-score data (traffic demand, network structure, field-based data) to diagnose inefficiencies in Bishkek's mobility system.
- Formulated mid- to long-term sustainable transport strategies integrating public transit enhancement and carbon reduction measures, proposing 190 initiatives across six strategic sectors.

Smart Transportation Lab, Myongji University

Seoul, Korea

Researcher

Mar. 2022 – Dec. 2024

"Development of Integrated Multimodal Mobility Service and Evaluation Platform"

National Research Foundation of Korea | Apr. 2024 – Dec. 2024

- Developed an integrated multimodal mobility platform with demand forecast and optimization models to support service planning and social benefit evaluation.
- Integrated a mobility service framework by linking data infrastructure with operational processes and analyzing first- and last-mile connectivity using smart card and bike-sharing data.

"Revision of the Korea Highway Capacity Manual (KHCM)" Korea Transport Institute (KOTI) | Jun. 2022 – Dec. 2024

- Revised KHCM to incorporate emerging mobility modes, evolving roadway system characteristics, and advanced traffic analysis methodologies affecting capacity and level of service.
- Developed and validated a conflict-based methodology using Surrogate Safety Measure (SSM), demonstrating that ramp influence areas extend beyond conventional speed-based definitions (~50m upstream).

"Study on Transfer Fare Integration and Periodic Revenue Clearing Mechanisms"

Metropolitan Transport Commission, Ministry of Land, Infrastructure and Transport (MOLIT) | Dec. 2023 – Jun. 2024

- Established a transparent and rational transfer fare clearing framework, including revised settlement rules, periodic clearing mechanisms, and inter-agency agreement schemes among metropolitan rail operators.
- Coordinated with 11 operators to align stakeholder interests and drive consensus on fare settlement mechanisms.

JOURNAL PUBLICATIONS

1. **Han, E.-S.**, Lee, E.-H., & Park, H.-C. (2026). "Evaluating Transfer Efficiency between Public Transportation and Bike Sharing for Mobility as a Service (MaaS)." *International Journal of Urban Sciences*, 1–24 (First Author).
2. Kim, K.-M., Jeon, J.-W., **Han, E.-S.**, & Park, H.-C. (2025). "A Data-driven Approach to Route Choice in Urban Railway Networks based on Gamma Distribution: A Case Study of the Seoul Metropolitan Area." *Transport (Accepted)*.
3. Lee, H.-S., Kwak, H.-C., **Han, E.-S.**, & Park, H.-C. (2024). "Modeling Urban Railway Passengers' Willingness to Wait based on Latent Class Analysis." *Transportation Research Record*, 2678(9), 230–240.
4. **Han, E.-S.**, & Park, H.-C. (2025). "Evaluation of Urban Rail Transfer Efficiency considering Intermodal Connectivity." *Journal of the Korean Society of Transportation*, 41(7), 771–778 (First Author).
5. **Han, E.-S.**, & Park, H.-C. (2024). "Applicability of Highway Entrance Ramp Influence Area using Surrogate Safety Measures based on NGSIM Data." *Traffic Safety Research*, 43(1), 71–89 (First Author).

CONFERENCE PROCEEDINGS

International (Refereed)

1. **Han, E.-S.**, Lee, H.-J., & Park, H.-C. (2025). "Evaluating Intermodal Transfer Efficiency between Public Transport and Bike-Sharing Systems in Seoul." *Transportation Research Symposium*, Rotterdam, Netherlands.
2. Lee, H., Kwak, H.-C., **Han, E.-S.**, & Park, H.-C. (2024). "Modeling Urban Railway Passengers' Willingness to Wait based on Latent Class Analysis." *TRB Annual Meeting*, Washington, DC, USA.

Domestic (Refereed)

1. **Han, E.-S.**, & Park, H.-C. (2025). "Efficiency Evaluation of Metropolitan Subway Transfer Stations Focusing on Integrated Transfers between Public Transportation Modes." *Korea Institute of ITS Spring Conference* [Oral Presentation, Best Paper Award].
2. **Han, E.-S.**, & Park, H.-C. (2025). "Efficiency Evaluation of Metropolitan Subway Transfer Stations Focusing on Integrated Transfers." *Suwon ITS Asia Pacific Forum*.
3. **Han, E.-S.**, & Park, H.-C. (2024). "Assessing the Suitability of Influence Area of Domestic Highway Entrance-Ramp based on Vehicle Trajectory Data." *Korea Institute of ITS Spring Conference*.
4. **Han, E.-S.**, & Park, H.-C. (2024). "DEA-based Evaluation of Subway Transfer Stations using Smart Card Data." *Korea Institute of ITS Fall Conference* [Oral Presentation, Best Paper Award].
5. **Han, E.-S.**, & Park, H.-C. (2023). "Estimating and Verification of Influence Area of Highway Entrance-Ramp through Surrogate Safety Measure (SSM) based on Vehicle Trajectory Data." *Korea Institute of ITS Fall Conference*.
6. **Han, E.-S.**, & Park, H.-C. (2023). "Methodology for Estimating Highway Entrance-Ramp Influence Area through SSM based on Vehicle Trajectory Data." *Korea Institute of ITS Spring Conference*.
7. **Han, E.-S.**, & Park, H.-C. (2022). "Analysis of Autonomous Vehicle Trends and Disengagement Causes based on California Reports." *Korea Institute of ITS Fall Conference*.

AWARDS & SCHOLARSHIPS

Best Paper Award, The 2025 Korea Institute of ITS Spring Conference	Apr. 2025
Best Paper Award, The 2024 Korea Institute of ITS Fall Conference	Oct. 2024
Grand Prize, Undergraduate Special Session, The 2023 Korea Institute of ITS Fall Conference	Apr. 2023
Grand Prize, Scholarship Selection Competition, Korean Society of Transportation	Feb. 2023
Outstanding First-Year Scholarship, Myongji University Graduate School	Aug. 2023 – Aug. 2024
Merit-based Scholarships, Myongji University (6 semesters)	Aug. 2021 – Aug. 2024
Welcome Financial Group Scholarship	Feb. 2022

SKILLS & PATENT

Technical Skills: SPSS, R, Python, QGIS, Vissim, AutoCAD, LaTeX, MS Office Suite (Word, PowerPoint, Excel)

Patent: Lee, H.-S., **Han, E.-S.**, & Park, H.-C., "Real-time Urban Railway Boarding Guidance System based on Passenger Behavior Prediction and Method Therefor," No. 10-2025-0097488 (2025), Republic of Korea.